

ORGAN SYSTEMS 2 - 2015
HUMAN GROSS ANATOMY AND EMBRYOLOGY
LEARNING OBJECTIVES

The most difficult aspect of studying anatomy is organizing your time to allow uptake and retention of a large quantity of information in a limited amount of time. To help you organize yourself, the following is a list of learning (study) objectives. The general objectives, listed first, apply to all course material. The specific structure objectives are important guides to pertinent topics in each of the integrated courses. Consider these carefully as you progress through the relevant dissections and lectures. Remember, all examination material will be derived from the study objectives.

GENERAL COURSE OBJECTIVES

1. Spell and pronounce anatomical terminology correctly and precisely.
2. Identify, differentiate and/or delimit the body regions, organ systems, organs, and subdivisions of the organs of the human body.
3. Describe the *in situ* relationships of the major structures in the human body.
4. Delineate the surface projections of clinically significant internal structures and their components.
5. Correlate structure and function of the major structures in the human body.
6. Integrate the gross and microscopic continuities of organs and organ systems within and among body regions.
7. Describe how organogenesis of systems accounts for the formation and disposition of organs in the adult body.
8. Describe the features of and developmental bases for the major congenital malformations of organs and regions.
9. Identify the major structures of the human body in standard medical images.
10. Predict the clinical significance and functional consequences of the major pathological conditions of normal human anatomy.
11. Identify the major structures of the human body in dissection.
12. Develop a three-dimensional working knowledge of the composition of the human body by the integration of its component parts.

SPECIFIC STRUCTURE OBJECTIVES - A SAMPLE PLAN

The following outline provides a precise method for organizing anatomical information. You may find this an effective study methodology. For each structure encountered in class, ask yourself the appropriate questions listed. Remember to use the outline only in the context of structures presented in dissection and lecture. Do not attempt to assimilate every structure noted in your textbook.

BONE

1. Location.
2. Articulation(s).
3. Developmental type (membrane or endochondral).
4. Growth centers (primary and secondary).
5. Maturation age.
6. Muscle, tendon, ligament attachments.
7. Neurovascular or other structures housed or conveyed or having close topographic relationship.
8. Surface projection and palpable points.
9. Effect of specific fractures.

JOINT

1. Type (classification and degrees of freedom permitted).
2. Bones involved (including specific parts of bones concerned).
3. Extent and attachment of capsule and lining.
4. Supporting extra-articular ligaments and muscles.
5. Intra-articular structures (articular disc and/or ligaments).
6. Movement caused by each muscle affecting joint.
7. Innervation.
8. Blood supply and drainage.

MUSCLE

1. Origin
2. Insertion.
3. Structure (subdivisions, fiber direction, pinnation, etc.).
4. Action (on which joint or joints - type of action).
5. Innervation.
6. Blood supply.
7. Relations.
8. Antagonist(s).
9. Effect of loss.
10. Surface projection and/or palpation.

BLOOD VESSEL

1. Artery or vein.
2. Origin (from what other vessel).
3. Course and relations (subdivide into parts if needed).
4. Named branches (where each originates).
5. Area supplied or drained by each named branch (give specific area).
6. If any branches have named divisions, list these.
7. Companion structures.
8. Major anastomoses and collateral routes.
9. Effect of blockage.
10. Surface projection and/or palpation.

NERVE

1. Functional components
2. Origin of fibers.
3. Course and relations of fibers.
4. Target(s) of fibers.
5. Location of cell bodies of fibers.
6. If autonomic (visceral) innervation:
 - a. Parasympathetic vs. Sympathetic.
 - b. Origin and course of preganglionic fibers.
 - c. Location of synapses.
 - d. Origin, course, and target of postganglionic fibers.
 - e. Afferent companions and projections.
7. Effect of loss.
8. Surface projection and/or palpation.

SPACE OR AREA

1. Location.
2. Boundaries (structures which make up boundaries).
3. Contents and/or transients.
4. Surface projection and/or palpation.

ORGAN

1. Definition (type).
2. Location.
3. Relations (contacts).
4. Structure (size, shape, lobes, fissures, ducts, stroma, parenchyma, etc.).
5. Function.
6. Blood supply and drainage.
7. Innervation.
8. Lymphatic drainage.
9. Developmental history.
10. Effect of loss or pathology.

11. Surface projection and/or palpation.

LYMPHATIC DRAINAGE

1. Area(s) drained.
 2. Course of channels from area drained to nodal groups.
 3. Names of node groups.
 4. Location of node groups.
 5. Routes of drainage from node groups to next major node groups.
 6. Drainage point with venous system.
 7. Palpation points of nodal groups.
 8. Function.
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POSTERIOR ABDOMINAL WALL

1. Identify the bones of the posterior abdominal wall and their major features. Describe the functional aspects of these structures.
 2. Identify the muscles of the posterior abdominal wall, noting their attachments, innervation, and major actions.
 3. Identify the nerves of the posterior abdominal wall. Predict the functional consequences of lesions to specific spinal levels and peripheral nerves.
 4. Identify the main arterial and venous branches to and from the posterior abdominal wall.
 5. Identify the kidneys and suprarenal glands. Describe their normal positions, relations, and neurovascular supplies.
 6. Trace the flow of urine from the gross collecting structures in the kidney to the urinary bladder. Note the relationships of the ureters to surrounding structures.
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PELVIS

1. Identify the bones of the pelvis and their major features. Describe the functional aspects of these structures. Identify these structures in standard medical imaging.
2. Identify the bony components, major supporting ligaments, key accessory structures (e.g., intra-articular discs), and movements permitted at the joints of the pelvis. Relate these to the general mechanics of the pelvis.
3. Identify the specialized pelvic differences between males and females. Relate these to the mechanics of the pelvis, including the construction of the birth canal.

4. Identify the pelvic diaphragm, its components, and their basic functions. Indicate differences between males and females with respect to relationships to the urogenital tracts.
 5. Identify the urinary bladder. Describe its gross features, relationship to the peritoneal cavity, position in the pelvis in both full and empty states, the nature and source of its innervation, and its means of filling and drainage in both males and females.
 6. Identify the rectum, anal canal, and the anal sphincters. Describe the relationships of these structures, with particular reference to the basis of a digital anorectal exam in both males and females. Describe the anatomical basis for control of defecation and cause of hemorrhoids.
 7. Trace the courses of the male and female reproductive tracts. Identify accessory glands or organs in the pathway. Note the relationships of components. Describe the nerves, muscles, and vasculature that are responsible for normal sexual functioning in males and females.
 8. Locate the surface projections and palpation points of major pelvic structures in a basic surface examination.
 9. Identify the major structures of the pelvis in radiographic and cross-sectional images.
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PERINEUM

1. Define and delineate the boundaries and subdivisions of the perineum. Identify the contents of each.
 2. Identify the urogenital diaphragm, its components, and their basic functions. Indicate differences between males and females with respect to relationships to the urogenital tracts.
 3. Delineate the ischiorectal (ischioanal) fossae. Identify the major contents of the fossae and describe the functional/clinical significance of the area.
 4. Describe the innervation, vascular supply and drainage, and lymphatic drainage of the perineum, with particular emphasis on urogenital and digestive tract structures.
 5. Describe the relations of the anorectal canal (in males and females) and the vagina as relevant to digital anorectal and vaginal examinations.
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NEUROVASCULAR ORGANIZATION OF THE PELVIS AND PERINEUM

1. Identify the sacral and coccygeal plexuses, their segmental origins, and their major peripheral branches. Predict the functional consequences of lesions to particular spinal levels and peripheral nerves.
 2. Trace the flow of blood into the pelvis and perineum by describing the courses, branching patterns, and distributions of the common iliac, external iliac, and internal iliac arteries. Note the relations of these vessels to neighboring organs, mesenteries, and major veins.
 3. Trace the flow of blood from the pelvis and perineum to the heart by describing the courses, branching patterns, and distributions of the iliac veins and inferior vena cava. Note the relations of the major veins along their paths.
 4. Trace the primary drainage routes of lymph from the organs and walls of the pelvic cavity and perineum to their primary lymph nodes and points of venous connection.
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DEVELOPMENT OF THE UROGENITAL SYSTEM

1. Describe the common origins of the urinary and genital systems, including the roles of the intermediate mesoderm and urogenital ridge.
2. Outline the development of the three sets of kidney systems in humans, with special emphasis on the development of the permanent kidneys. Note the time sequence of formation, functional capabilities, and contributions to adult form of each. Trace the formation of the collecting and excretory components and the positional changes of the permanent kidney. Trace the path of urine flow from the point of production within the nephron to the urinary bladder.
3. Describe the development of the urinary bladder and urethra, relating these to simultaneous development of the kidney and reproductive tracts. Trace the path of urine excretion from the urinary bladder to the exterior in males and females.
4. Describe the characteristics and bases of the major congenital defects in urinary tract development.
5. Outline the process of sex determination in human embryos. Include descriptions of:
 - a. The appearance, migration, and significance of primordial germ cells.
 - b. The differentiation and roles of definitive gonads.
 - c. The difference between genetic and phenotypic sex.
6. Compare and contrast the development of the genital duct systems and external genitalia in males and females. Note the fate of the mesonephric and paramesonephric ducts and the development of the seminal and prostate glands, uterus and vagina, and urethra. List

the homologies between male and female definitive adult structures and clinically significant vestigial structures.

5. Describe the characteristics and bases of the major congenital defects in reproductive tract development.

ORGANIZATION OF THE LOWER LIMB

1. Identify the bones of the lower limb and their major features. Describe the functional aspects of these structures. Identify these structures in standard medical imaging.
2. Compare and contrast the pattern of development of the lower limb with that of the upper limb.
3. Identify the lumbosacral plexus and its component parts, from spinal segmental sources to major terminal branches.
4. Identify the muscles of the gluteal region, indicating their attachments, innervation, and major actions. Note the roles of the gluteal muscles during locomotion.
5. Describe the geographic relationships of neurovascular structures in the gluteal region and the consequences of intragluteal injections into specific quadrants of the region.
6. Describe the arrangement and mechanical significance of the deep fascia in the lower limb. Identify the fascia lata, iliotibial tract, and intermuscular septa in the hip and thigh.
7. Define the osseofascial compartments of the lower limb.
 - a. Identify the muscles contained in each compartment.
 - b. Describe the attachments, innervation, and major actions of each muscle.
 - c. Describe the innervation and major actions of each compartment as a whole.
 - d. Predict the functional consequences of loss of action of each muscle and each compartment.
8. Trace the course of cutaneous and motor innervation in the gluteal region and lower limb.
 - a. Identify the spinal segmental level(s) of origin and relationship to the lumbosacral plexus of each major peripheral nerve.
 - b. Predict the functional consequences of lesions to specific spinal levels and individual peripheral nerves.
9. Define the boundaries and contents of the femoral triangle. Describe the composition of the femoral sheath, canal, and ring. Describe the anatomical basis of a femoral hernia.
10. Define the boundaries of the popliteal fossa and describe its contents. Note the spatial

relationships of the major neurovascular structures in the fossa. Identify the main components of the arterial anastomoses around the knee.

11. Trace the flow of blood from the common iliac artery to and through the gluteal region and lower limb. Note:
 - a. The different sources of arterial supply, their pathways, and their major branches.
 - b. The significance of the obturator canal, femoral sheath, adductor canal, adductor hiatus, greater sciatic foramen, and lesser sciatic foramen.
 - c. The composition and significance of the cruciate anastomosis.
 - d. The territories supplied by and the pathways of the major peripheral vessels.
 12. Trace the venous drainage of the lower limb through the saphenous tract to the pelvic cavity. Describe the significance of the saphenous veins in terms of:
 - a. Varicose veins.
 - b. Coronary bypass surgery.
 13. Describe the pattern of lymphatic drainage of the lower limb, including the relationship of this drainage with those of the abdominal wall and groin regions.
 14. Determine the movements that occur at the knee joint during normal locomotion. Describe the mechanics of "locking" and "unlocking" of the knee.
 15. Identify the extrinsic and intrinsic muscles of the foot, indicating their attachments, innervation, and major actions. Compare and contrast the patterning and functions of muscles of the hand and foot.
 16. Describe the arrangements of flexor, extensor, and peroneal retinacula, and major tendons around the ankle, including the functional significance of these arrangements.
 17. Identify the bony components, major supporting ligaments, key accessory structures (e.g., intraarticular discs), and movements permitted at the hip, knee, and ankle joints. Describe the features of major traumas to each joint.
 18. Identify the medial and lateral longitudinal and transverse arches of the foot. Describe the roles of bones, ligaments, and muscles in maintaining these arches.
 19. Locate the projections and palpation points of the major structures of the lower limb in a basic surface examination.
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